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Globular Cluster Classification Using Machine Learning on Fornax Deep Survey and *Euclid* Data on the Fornax Cluster

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ABSTRACT

Globular Clusters (GCs) are crucial objects to identify in galaxies far away in order to learn more about the galaxies they are part of. Because of their point-like appearance, they can be hard to distinguish. In this work, we apply a variety of machine learning (ML) methods to both Fornax and *Euclid* datasets in order to try to find these objects. In our best method of using a k-Nearest Neighbor (kNN) classifier, we found a recall/precision score of of .88/.84 for Fornax, and .93/.86 for *Euclid*. These results build on the ever-growing evidence that ML techniques are a vital technique and will continue to be for wide-field surveys such as the LSST of Rubin.

1. INTRODUCTION

Globular Clusters (GCs) are dense, spherical systems of ancient stars that serve as powerful observational tools in our extragalactic understanding. Because they form during major starburst events and survive over long timescales, GCs can act as fossil records, tracing the assembly histories of their host galaxies. Because of this, being able to accurately catalog all known GCs is an important step in continuing our undersanding of galaxy evolution.

However, identifying GCs at large distances presents a significant observational challenge. Due to their compact nature, GCs suffer from visual overlap with other astronomical sources. Specifically, they are difficult to distinguish from foreground Milky Way stars, as well as unresolved background galaxies. Foreground stars can appear point-like, and background galaxies may appear to be only slightly extended. Traditionally, manual separation by making cuts in color has had some success but may prone to high contamination rates.

Machine Learning (ML) techniques are a powerful tool to both automate and improve upon this process. Previous includes Barbisan et al. (2022) on M87 and Floyd et al. (2024) work on nearby spirals and their GC systems. Work has also been done on the Fornax cluster on separating GCs from objects using various ML approaches Saifollahi et al. (2021); Ordenes-Briceño et al. (2026). Methods have even been applied to simulated data to prepare for the wealth of data that the Rubin Observatory will provide Schweder-Souza et al. (2025).

In this paper, we explore the application of several ML techniques to accurately identify GCs. We apply these methods to data from the Fornax cluster using the Fornax Deep Survey (FDS) and *Euclid*, and compare their performance to determine the best method for GC candidate selection.

2. METHODS

2.1. *Data Exploration*

The initial FDS dataset has all three classes of objects separated already. There are 648 GCs, 4166 stars, and 2151 galaxies. This dataset is unbalanced, so it will be something to look out for when applying my methods. This dataset includes coordinates of RA and Dec alongside it, and is a feature to consider leveraging in our models.

The *Euclid* dataset is also pre-separated, with 121 GCs, 452 stars, and 172 galaxies. This has a higher percentage of GCs relative to the Fornax data, although a much smaller sample size of objects to work with. Histograms of the data can be seen in Figure 1, showing the large amount of galaxies and especially stars to GCs. We can also see the difference in brightness between GCs and the other two objects, when looking at the central peaks of both histograms.

Cornerplots of a subsuet offeatures may be seen in Figures 2 and 3. Cornerplots using PCA components for both datasets may be found in the appendix A.

2.2. *Model Methods*

As done in previous literature (Schweder-Souza et al. 2025; Saifollahi et al. 2021; Ordenes-Briceño et al. 2026), I have feature-engineered colors using these bands, and then I chose to use the Robust Scaler on my color columns.

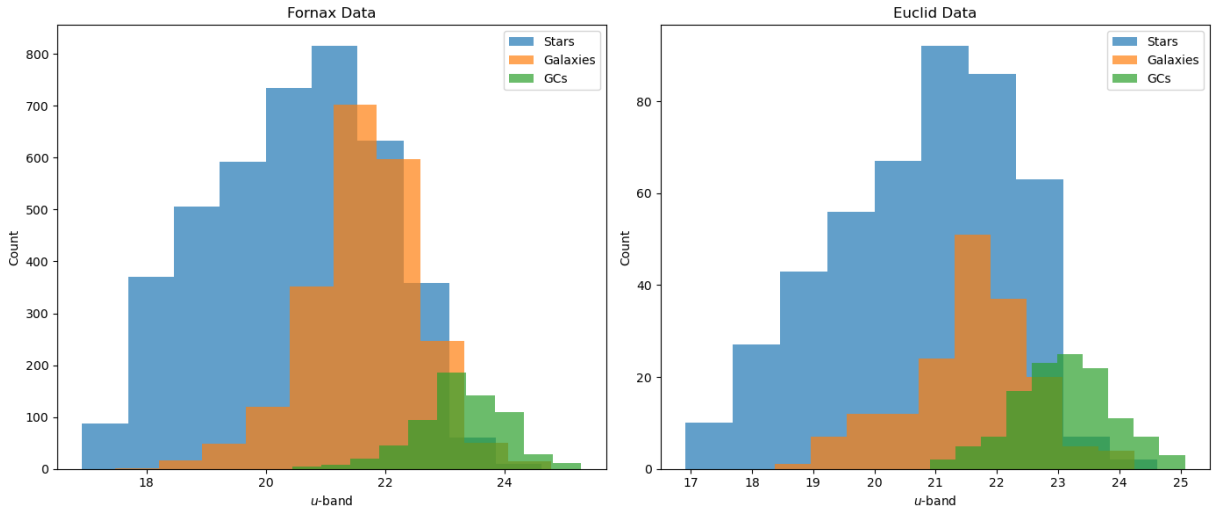


Figure 1. Initial data visualization of both datasets using the u-band, before applying any scaling.

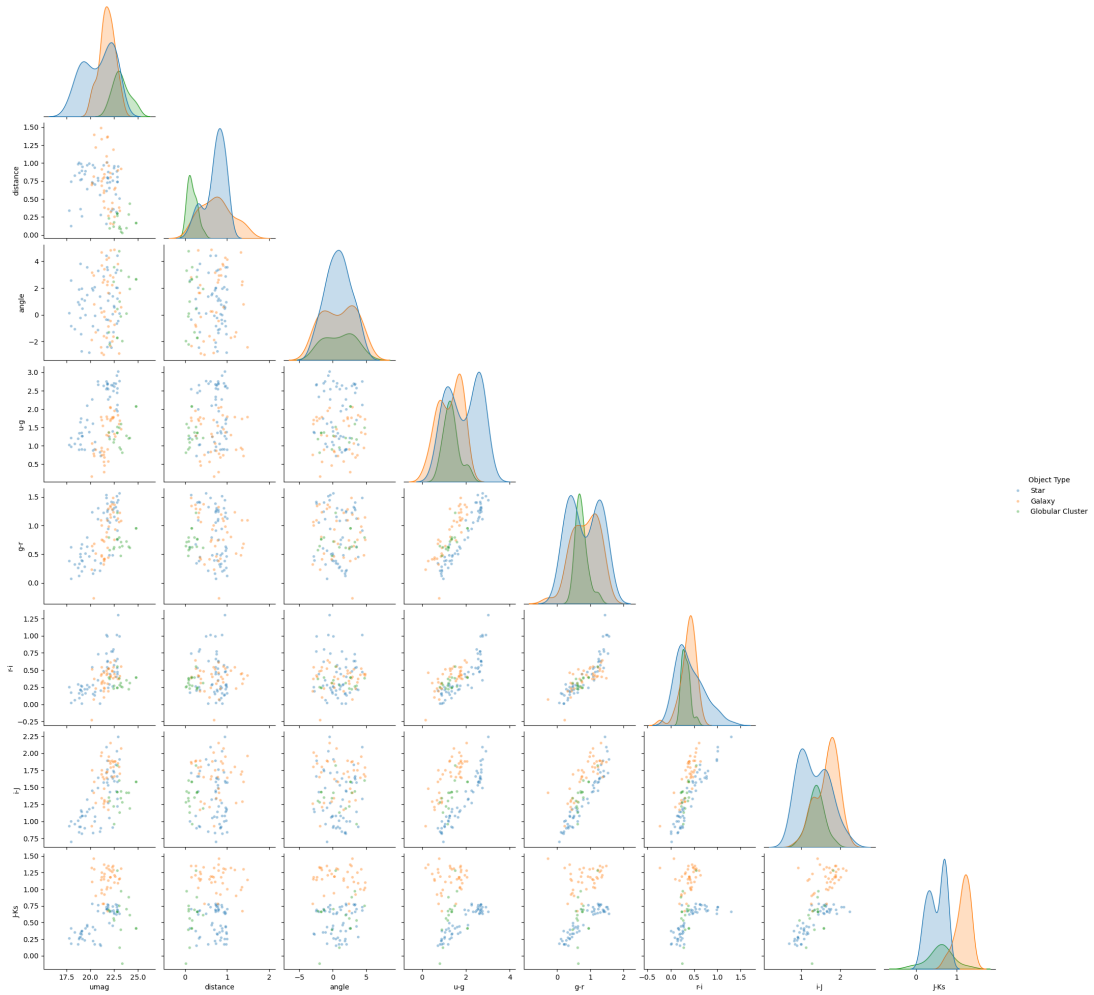


Figure 2. Cornerplot of a subset of Fornax color columns.

Scaling is a crucial part in many ML techniques to keep relative differences important for any feature. Initially, I tried

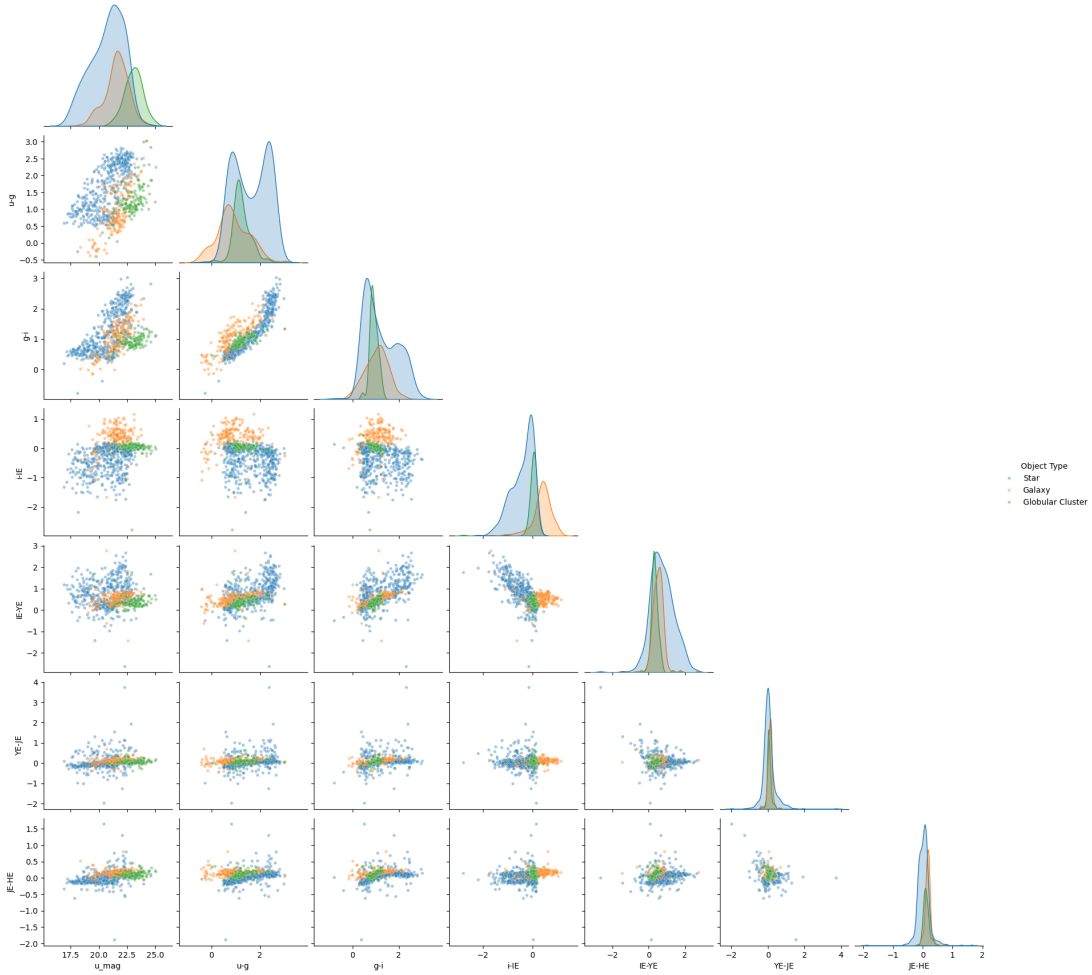


Figure 3. Cornerplot of a subset of *Euclid* color columns.

using only the exact 5 color features from Saifollahi et al. (2021) in the Fornax dataset for comparison. Then, I kept the *u*mag column, because keeping one photometry band on its own helps performance. Including the RA and Dec information of the Fornax data did not yield any increases in score. However, feature engineering a distance and angle from these RA and Dec columns from the center of the Fornax Cluster did seem to increase scores slightly. These were excluded in order to be independent from those features. This results in a total of 16 features in the Fornax dataset, and 22 features in the *Euclid* dataset.

I ran a dummy classifier on the datasets beforehand, choosing the "stratified" option, which chooses a random classification weighted by the labels of the dataset. For PCA and each model, I used the scikit-learn package. I then used principal-component analysis (PCA) on my columns for both sets of data before running models. For both datasets, using a number of PCA components up to 10 increased scores, with no increase in scores when going past 10. Using the 10 PCA components, I first used the RFC using n=100 estimators. I did cross validation, with cv= 5folds.

The second model included an MLP model. for both datasets. For this I chose to have 2 hidden layers of 10 neurons each. I used 1000 epochs and a learning rate of .01. I ran a k-nearest neighbors (KNN) model to compare with Saifollahi et al. (2021) results. For this I used $n = 5$ neighbors for both datasets.

The final model was a SVM model. I chose to use an rbf (gaussian) kernel and run a grid search over γ and C values. I used γ= 0.01, 0.1, 1, 10 and C=0.1, 1, 10, 100 for both datasets. I chose to score using recall-macro. Both datasets scored the best with C=10, and with γ= 0.1 for Fornax data, and γ=0.01 for the *Euclid* dataset. the class weight parameter was set to "balanced."

3. RESULTS

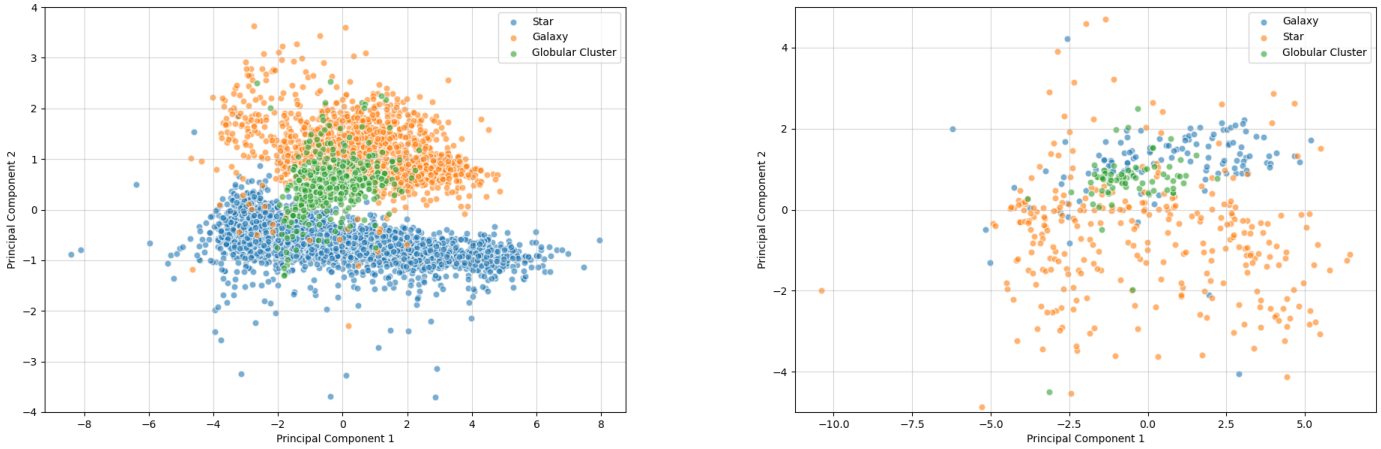


Figure 4. First 2 PCA components for both datasets. Left side is Fornax, and the right side is *Euclid*.

The initial model using just 5 color bands on a kNN model as done in Saifollahi et al. (2021) underperformed their recall value, but surpassed their precision value. These values can be seen in table 3.

Using all features, the kNN model performed the best without using PCA components. Using a stratified dummy model gave recall of 0.13. These values can be seen in Table 3. Results of each different model can be seen in appendix A

Class	Precision	Recall	F1-score	Support
Stars	0.98	0.98	0.98	4166
GCs	0.84	0.88	0.86	648
Galaxies	0.96	0.96	0.96	2151
Accuracy	-	-	0.96	6965
Macro avg	0.93	0.94	0.94	6965
Weighted avg	0.96	0.96	0.96	6965

Table 1. Fornax kNN results

The kNN model also worked the best for *Euclid* data without PCA components, although was close to the RF model. These values can be seen in Table 3. The same stratified dummy model gave a recall value of 0.12. Results of each model can be seen in appendix A.

Class	Precision	Recall	F1-score	Support
Stars	0.96	0.96	0.96	452
GCs	0.86	0.93	0.89	121
Galaxies	0.97	.90	0.93	173
Accuracy	-	-	0.98	746
Macro avg	0.98	0.98	0.98	746
Weighted avg	0.98	0.98	0.98	746

Table 2. *Euclid* kNN results

4. DISCUSSION/CONCLUSION

Class	Precision	Recall	F1-score	Support
Stars	0.97	0.99	0.98	736
GCs	0.84	0.73	0.78	198
Galaxies	0.96	0.96	0.96	1365
Accuracy	-	-	0.95	2299
Macro avg	0.92	0.89	0.91	2299
Weighted avg	0.95	0.95	0.95	2299

Table 3. Fornax kNN results, using only 5 colorbands from Saifollahi et al. (2021)

Work by Saifollahi et al. (2021) using only 5 colors has a recall/precison rate of .81/.77 compared to my kNN precision/recall of .73/.84 (seen in Table 3). My model produces a higher precision, but worse recall. Because recall is a more important metric, to improve my 5-color model I would need to see if there is more data preparation I can do in terms of removing bad candidates before starting my models. They used a neighbor value of k=7, although my best value was at k=5.

Learning Curves can be found in Figures 9 and 10. My RF model shows a relatively high variance with both datasets. The SVM model appears to continue to be converging between the training and test scores, so more data could assist that model. These show variations in the bias and variance in the data. The most important features are listed in Figures 5 and 6. In both cases, a color is the defining feature, with the u-band magnitude being behind it as the second most important. Towards the bottom are colors that carry little information due to some redundancy and overlap between some of them.

Using PCA components slightly decreased the performance of most models, with it affecting the kNN models the least. Future work includes analyzing in what scenarios the computation time save of PCA components would be worth using over the small drop in accuracy of the models. If it fits within the science goals, then RA/Dec information may be leveraged for small performance increases for a dataset such as the Fornax cluster. There is some amount information in where these GCs lie relative to the center of this cluster of galaxies.

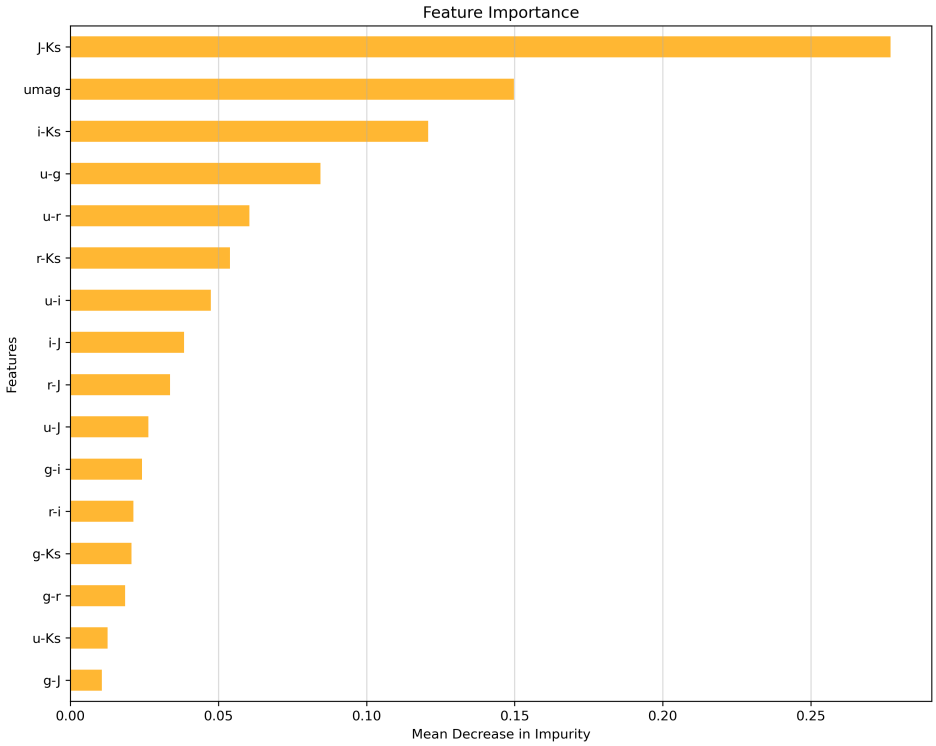


Figure 5. Feature Importance of Fornax data.

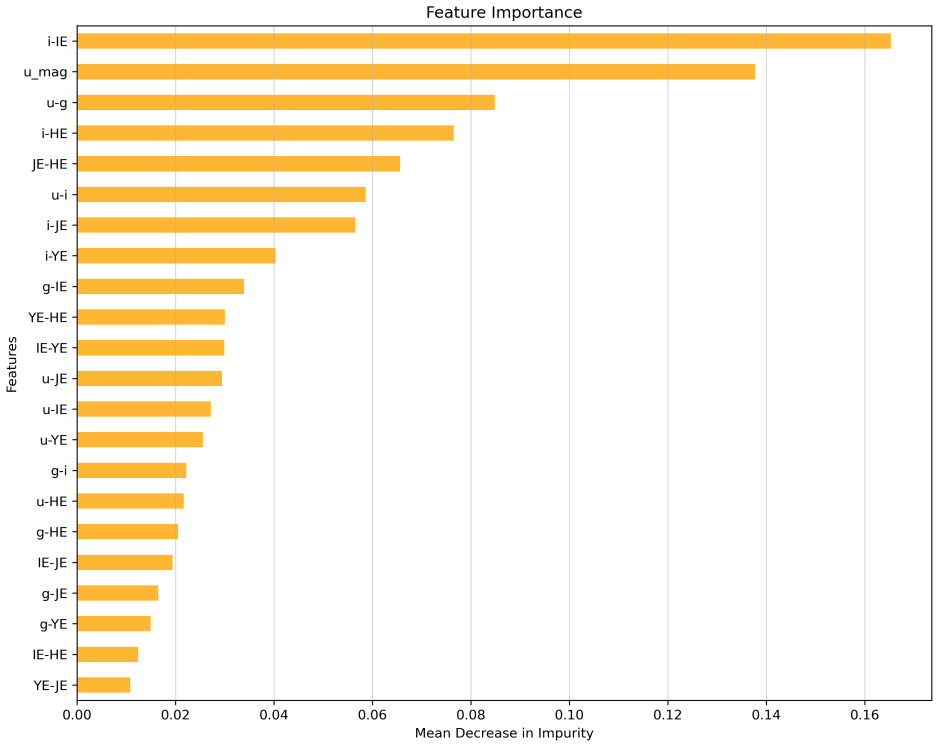


Figure 6. Feature Importance of *Euclid* data.

A. APPENDIX

Here are relevant figures, put at the end of this paper to ease the reading pace above. There includes tables of all model results (Tables A,A), PCA component cornerplots (Figures 7,8), and learning curves for all 4 models (Figures 9 and 10).

Model	Precision	Recall	F1-score
RF	0.86	0.79	0.82
MLP	0.82	0.90	0.81
kNN	0.86	.93	0.89
SVM	0.82	0.91	0.86

Table 4. Fornax: All model results for GC classification

REFERENCES

Model	Precision	Recall	F1-score
RF	0.82	0.80	0.81
MLP	0.82	0.89	0.85
kNN	0.86	.93	0.89
SVM	0.82	0.91	0.86

Table 5. *Euclid*: All model results for GC classification

Barbisan, E., Huang, J., Dage, K. C., et al. 2022, MNRAS, 514, 943, doi: 10.1093/mnras/stac1396

Floyd, M., Chandar, R., Whitmore, B. C., et al. 2024, AJ, 167, 95, doi: 10.3847/1538-3881/ad1889

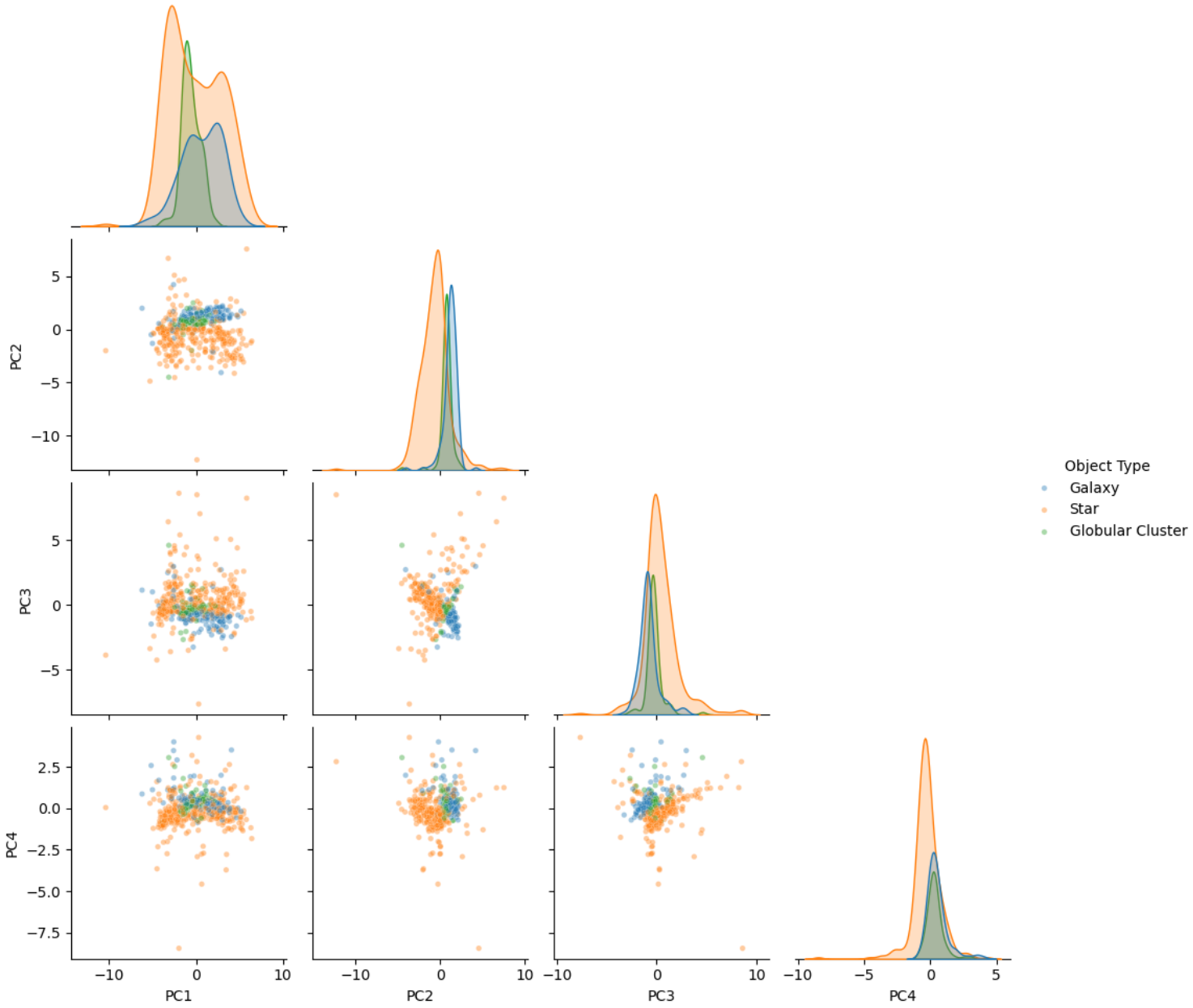


Figure 7. Pairplot of first four Fornax PCA components.

Ordenes-Briceño, Y., Puzia, T. H., Eigenthaler, P., et al. 2026, A&A, 706, A368, doi: 10.1051/0004-6361/202556455

Saifollahi, T., Janz, J., Peletier, R. F., et al. 2021, MNRAS, 504, 3580, doi: 10.1093/mnras/stab1118

Schweder-Souza, N., Chies-Santos, A. L., de Souza, R. S., et al. 2025, arXiv e-prints, arXiv:2512.17644, doi: 10.48550/arXiv.2512.17644

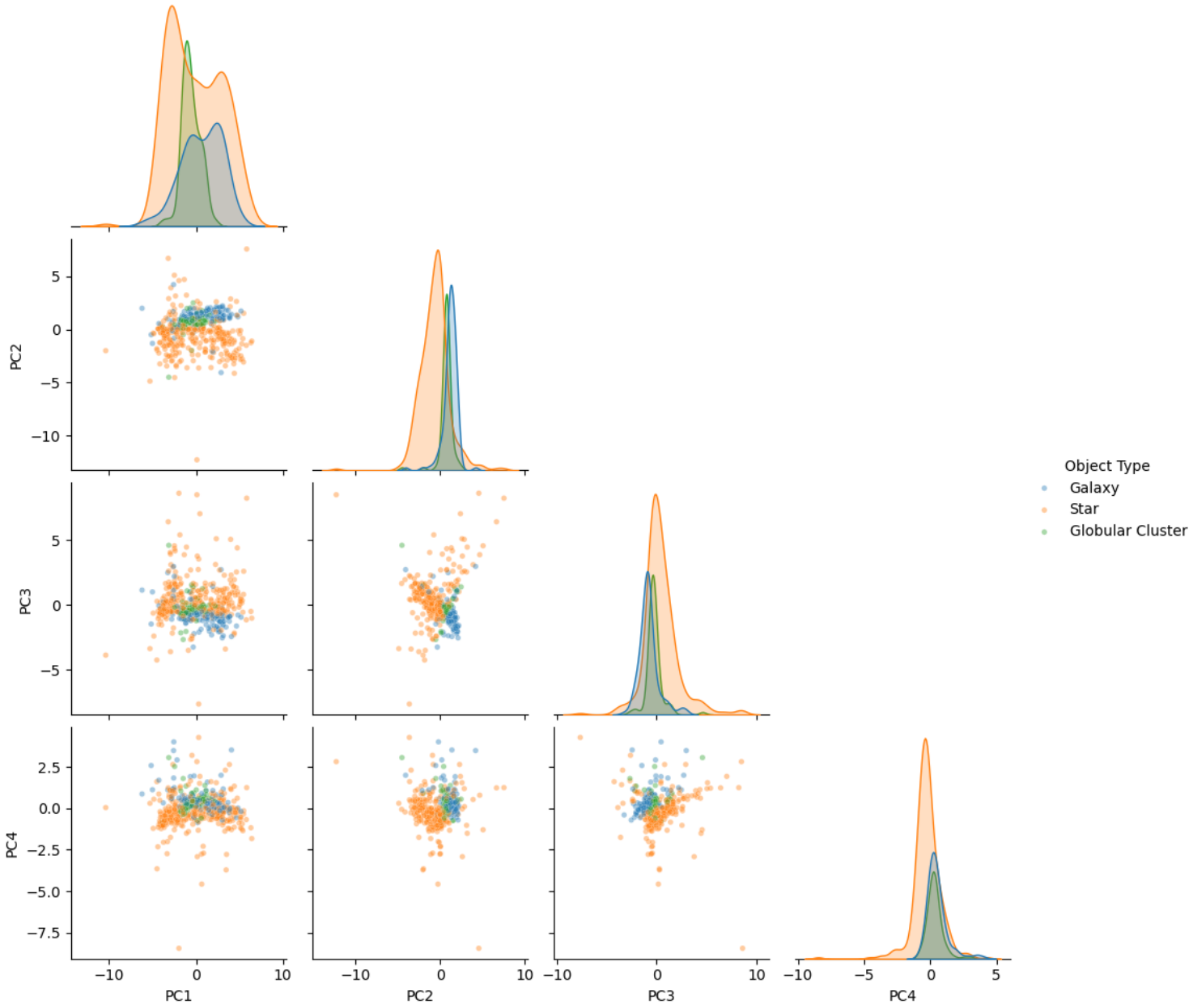


Figure 8. Pairplot of first four *Euclid* PCA components.

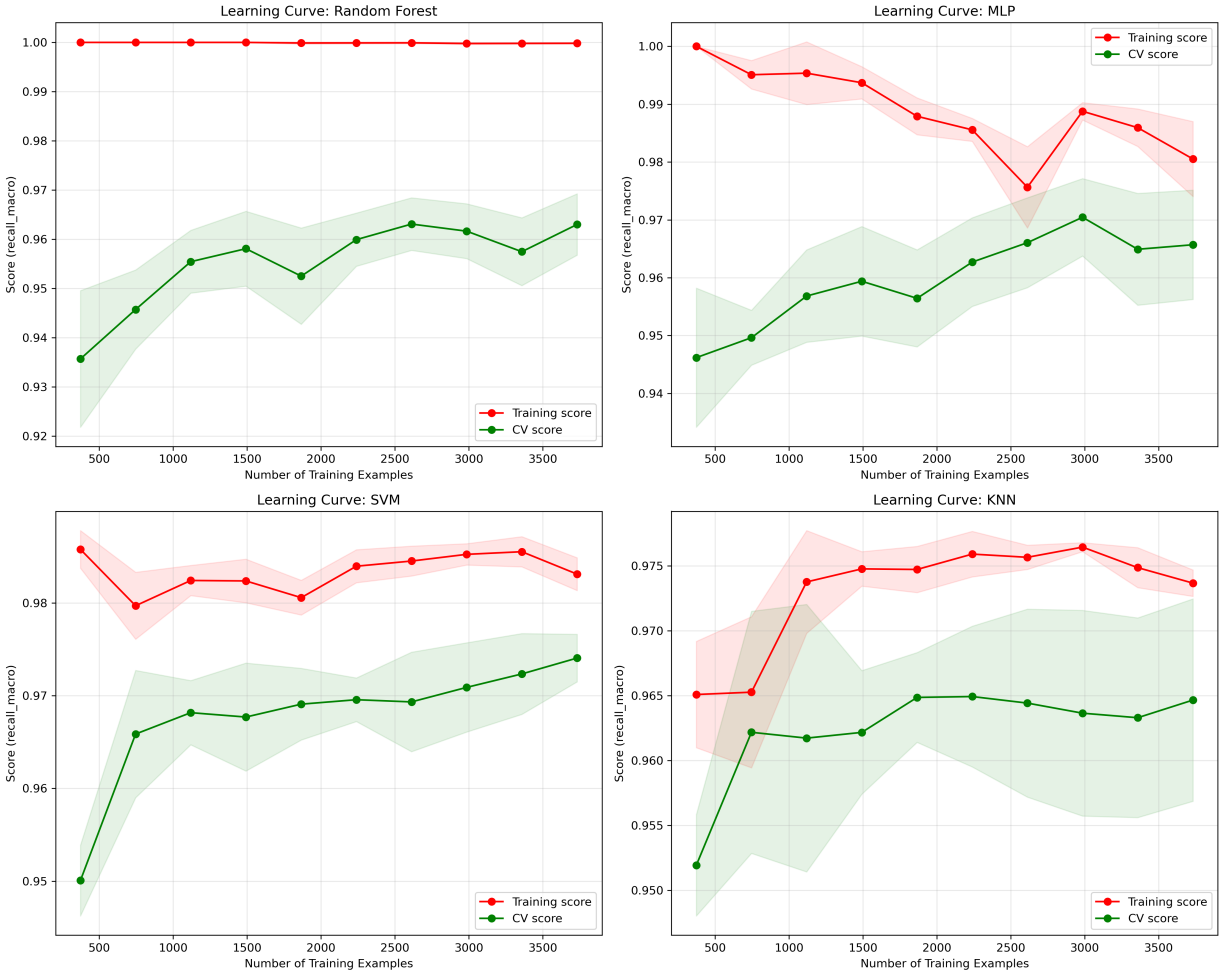


Figure 9. Learning Curve of all models for Fornax data.

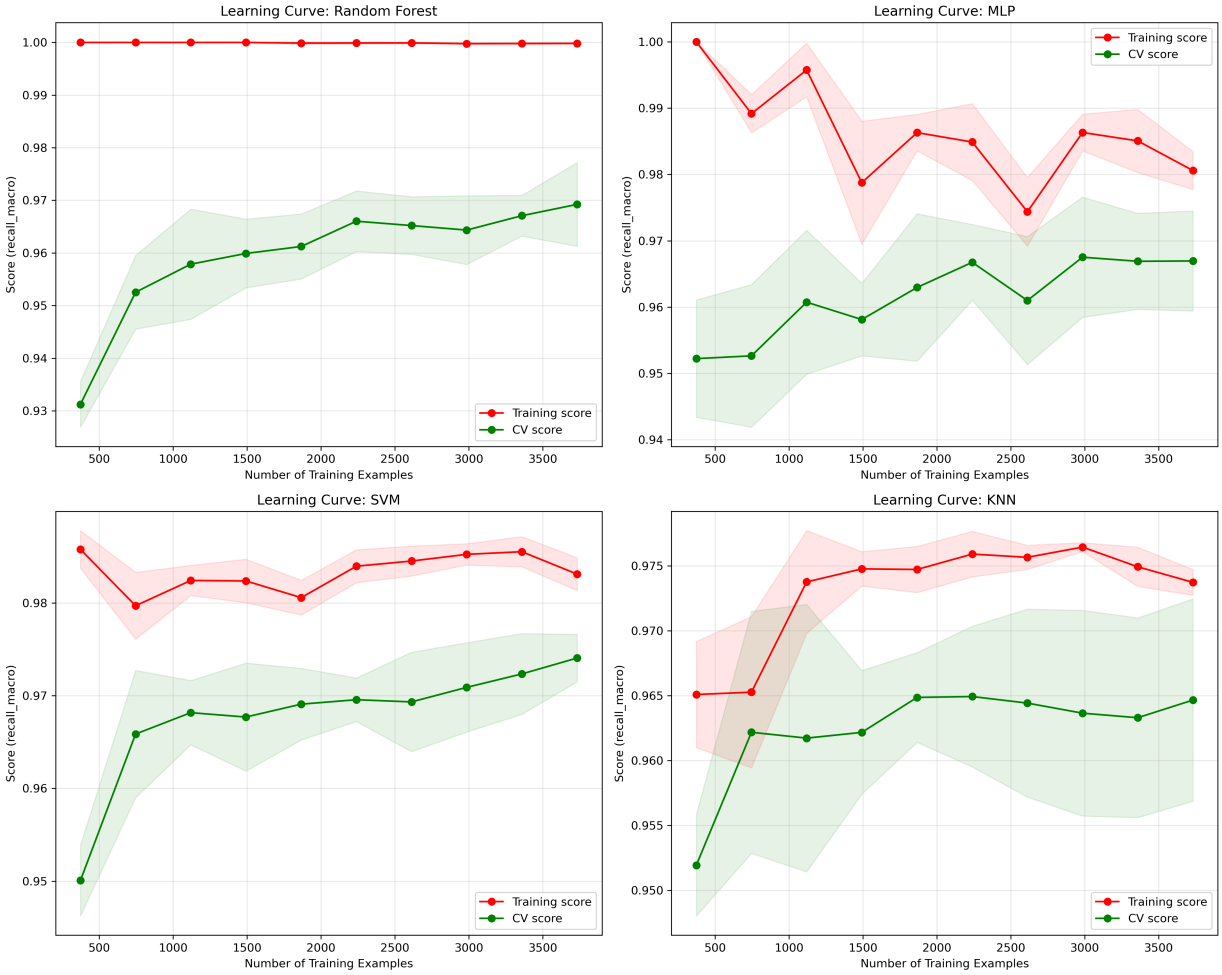


Figure 10. Learning Curve of all models for *Euclid* data.